Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

SEMESTER 1 EXAMINATIONS 2019/2020

MODULE: CA119 - Data Science and Databases

PROGRAMME(S):
DS BSc in Data Science

YEAR OF STUDY: 1

EXAMINER(S):
Mark Roantree (Internal) (Ext:5636)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer 4 questions. All questions carry equal marks.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 (SQL Programming) [TOTAL MARKS: 25]

This question uses the schema in Appendix A

Q 1(a) *[4 Marks]*
Get name, hire date, salary, and department number for those employees whose first name does not containing the letter T.
Sort the results in ascending order by department number.

Q 1(b) *[4 Marks]*
Display the name of all employees and the name of their manager.

Q 1(c) *[6 Marks]*
Display the employee number, name, job ID and salary for all employees whose salary is smaller than any salary of those employees whose job title is MK_MAN.
Exclude employees who have the Job ID ‘MK_MAN’ from your results.

Q 1(d) *[5 Marks]*
Display the department name, average salary and number of employees working in their department who got commission.

Q 1(e) *[6 Marks]*
Display job ID, number of employees, sum of salary, and difference between highest salary and lowest salary for each job.

[End of Question 1]

QUESTION 2 (The Relational Model) [TOTAL MARKS: 25]

Q 2(a) *[5 Marks]*

- What is the cardinality of the relation in Appendix B?
- What is the degree of the relation in Appendix B?
- Which of the above is related to the intension of the relation? Explain why you gave this answer.

Q 2(b) *[8 Marks]*

- Write the formula for the cartesian product of the 2 tables *R* and *S* in Appendices B and C.
- Write the result of this expression to your exam paper.

Q 2(c) *[5 Marks]*
Define what is meant by a *Domain* constraint. Provide an example of breaking this constraint for the schema in Appendix B.

Q 2(d) *[4 Marks]*
List what you believe to be all superkeys for the relation STUDENTS in Appendix B.

Q 2(e) *[3 Marks]*
Using the relation STUDENTS, provide an example of a violation of the *delete constraint*. Explain how the violation occurs.

[End of Question 2]

QUESTION 3 (Functional Dependencies) **[TOTAL MARKS: 25]**

Q 3(a) *[10 Marks]*
Assuming that a functional dependency is of the form L→ R where L represents one or more (determinant) attributes and R a different set of attributes, list 5 functional dependencies for the STUDENTS relation in Appendix B.
Provide a *different* value for L in each dependency.
(Hint: there are many possibilities).

Q 3(b) *[4 Marks]*
Provide an example of an *insertion anomaly* for the relation STUDENTS (appendix B) and the relation PROFESSORS (appendix C).

Q 3(c) *[6 Marks]*
What is meant by the *dependency preservation* property? In your answer, explain the circumstances in which it may occur and how it should be resolved.

Q 3(d) *[5 Marks]*
Assume you have a relation R with attributes {a,b,c,d,e}. Briefly describe an algorithm to locate functional dependencies in R.

[End of Question 3]

QUESTION 4 (Normalisation) **[TOTAL MARKS: 25]**

Q 4(a) *[6 Marks]*

- In what normal form would you consider the relation STUDENTS (in appendix B) to be? Why?
- In what normal form would you consider the relation PROFESSORS (in appendix C) to be? Why?

Q 4(b) *[13 Marks]*

Taking the relation STUDENTS, explain the steps you would take to bring this relation to third normal form. At each step, highlight the specific functional dependency that controlled your decomposition.

Q 4(c) *[6 Marks]*

Take the relation PROFESSORS, and create an unnormalised form of the relation to your answer book. Explain what you did and why it became an unnormalised relation.

[End of Question 4]

QUESTION 5 (Entity-Relationship Modelling) **[TOTAL MARKS: 25]**

Consider an E-R diagram for an airport management system with 4 entities: **Airline**, **Passenger**, **Flight** and **BoardingCard**.

Q 5(a) *[7 Marks]*

Name 3 simple attributes for each entity.
Name and describe 1 complex entity each for Passenger and Airline.
Name and describe 1 multi-valued entity each for Flight and Passenger.

Q 5(b) *[8 Marks]*

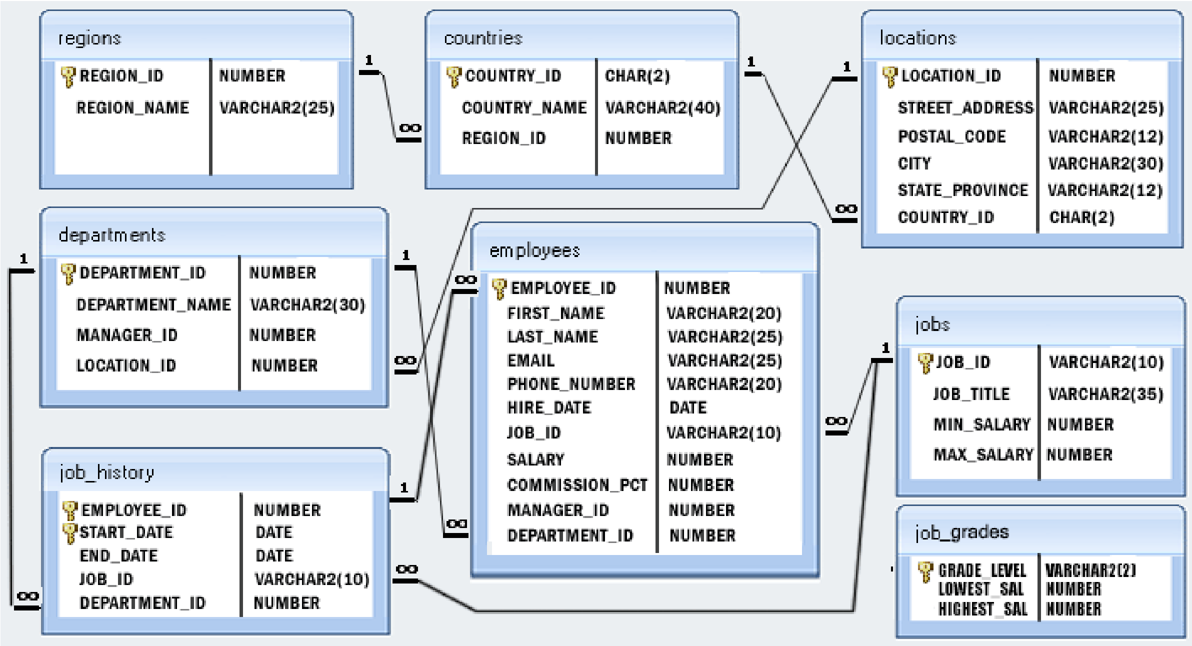
Which entities have a 1-1 relationship between them? Explain your answer.
Which entities have a 1-N relationship between them? Explain your answer.
Is there a M-N relationship in this E-R diagram? Explain your answer.

Q 5(c) *[10 Marks]*

Draw the final E-R diagram that corresponds to the answers you have provided in parts a) and b).

[End of Question 5]

Appendix A



HR Department Schema

Appendix B

SID	Name	Address	CourseID	CourseName	TutorID	TutorName
20181011	Michael Murphy	Dundalk	DC101	Computer Applications	750001	Jack Sharp
20181015	Ann Burke	Dunboyne	DC101	Computer Applications	750003	Pauline Burke
20181022	Tom Flynn	Finglas	DC101	Computer Applications	750004	Nora Martin
20181111	Mary Burke	Glasnevin	DC101	Computer Applications	750004	Nora Martin
20181211	Brian Byrne	Glasnevin	DC102	Enterprise Computing	750004	Nora Martin
20181256	Ellen Molloy	Glasnevin	DC102	Enterprise Computing	750007	Eilis Flynn
20181260	Deirdre flynn	Finglas	DC101	Computer Applications	750007	Eilis Flynn
20181275	Alan Marr	Finglas	DC102	Enterprise Computing	750001	Jack Sharp

Table B: Relation R *or* STUDENTS relation

Appendix C

CourseID	Department	Head
DC101	Department of Computing	Prof. Murphy
DC102	Department of Biology	Prof. Swan
DC103	Department of Physics	Prof. McDonald

Table C: Relation S *or* PROFESSORS relation

[END OF EXAM]